

McGill University
ECSE 222 – Digital Logic (Fall 2025)
Pen-and-Paper Assignment #3

Instructions

- TA in charge: Abidur Rahman (abidur.rahman@mail.mcgill.ca)
- Write/type your answers in the space provided.
- Due date: Please refer to myCourses.
- You must electronically submit the assignment in the form of a PDF file (only this format will be accepted) on myCourses before the deadline. You may submit as many times as you want, however, only the last submission will be kept by the system.
- You must solve the assignment in teams of two students, whereas only one of the students should perform the actual electronic submission.
- Late submissions will incur penalties as described in the course syllabus.

Student identification

First student full name:	
First student ID number:	
Second student full name:	
Second student ID number:	
Third student full name (if applicable)	
Third student ID number (if applicable)	

Evaluation (for course graders use only)

Question 1 (25%)	Question 2 (25%)	Question 3 (20%)	Question 4 (15%)	Question 5 (15%)	Total

Pen and Paper Assignment #3

Alexandre PUSAL

PLG.CC-ID: 261183554

Question 1:

We have $S = \begin{cases} A+B & \text{if } A \text{ and } B \text{ odd} \\ A & \text{if } A \text{ odd and } B \text{ even} \\ B & \text{if } B \text{ odd and } A \text{ even} \\ 0 & \text{if } A \text{ and } B \text{ even} \end{cases}$

$A = \{a_3 a_2 a_1 a_0\}$ $B = \{b_3 b_2 b_1 b_0\}$

Case 1: $S = A+B$

$S = \begin{matrix} & c_3 & c_2 & c_1 & c_0 \\ a_3 & a_2 & a_1 & a_0 \\ + & b_3 & b_2 & b_1 & b_0 \end{matrix}$

Truth Tables:

c_0	a_0	b_0	c_1	s_0
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0

since $c_0 = 0$
always

$c_1 = a_0 \oplus b_0$ $s_0 = a_0 \oplus b_0$

c_i	a_i	b_i	c_{i+1}	s_i
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

$\forall i \in \{1, 2, 3\}$

$c_{i+1} = a_i \oplus b_i$ $s_i = a_i \oplus b_i$

We want to implement all the different cases together

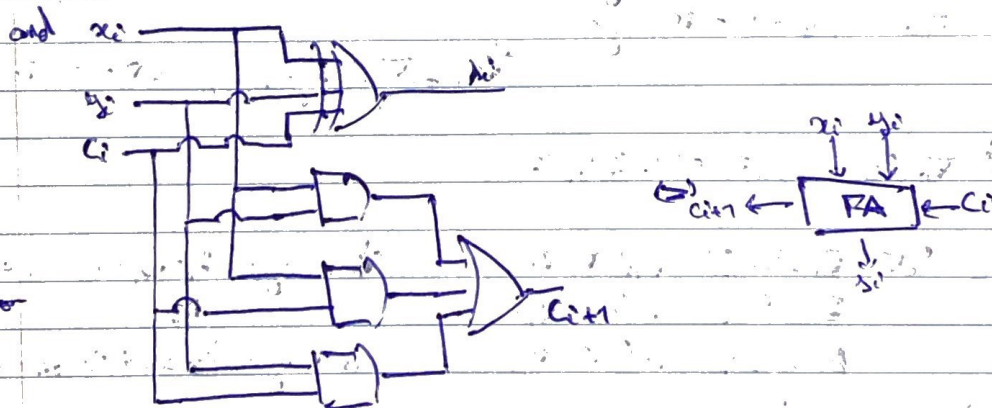
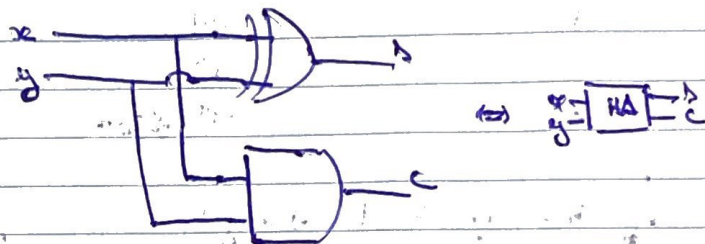
A is odd iff $a_0 = 1$ and even iff $a_0 = 0$
B is odd iff $b_0 = 1$ and even iff $b_0 = 0$

let $X = \{x_3 x_2 x_1 x_0\}$ with $x_i = a_i$ iff $a_i = 1$ and $Y = \{y_3 y_2 y_1 y_0\}$ with $y_i = b_i$ iff $b_i = 1$

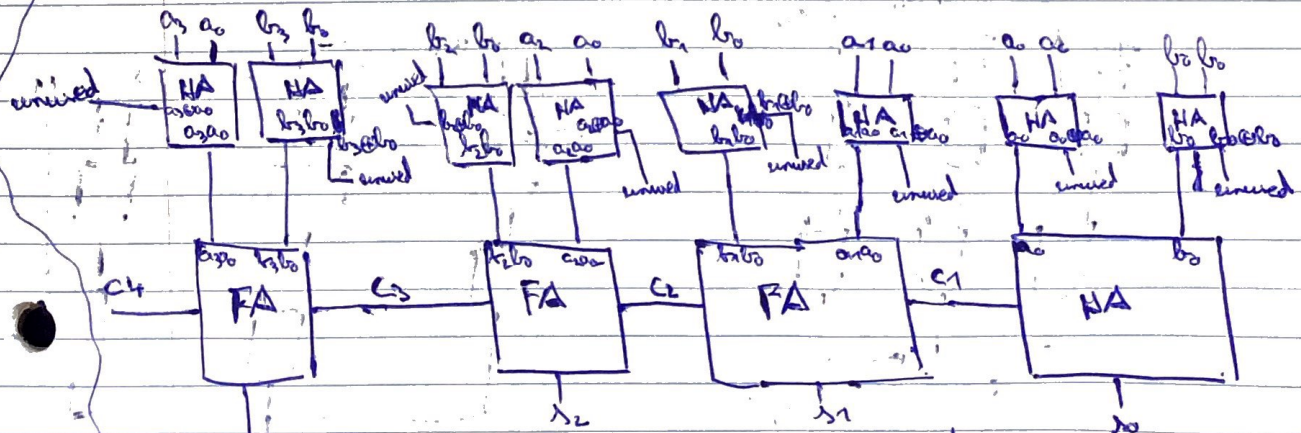
$$\text{and } S \equiv X+Y = \begin{cases} X+Y, & a_0=1 \text{ and } b_0=1 \Leftrightarrow \text{A and B odd} \\ X, & a_0=1 \text{ and } b_0=0 \Leftrightarrow \text{A odd and B even} \\ Y, & a_0=0 \text{ and } b_0=1 \Leftrightarrow \text{A even and B odd} \\ 0, & a_0=0 \text{ and } b_0=0 \Leftrightarrow \text{A and B even} \end{cases}$$

$$\text{Hence } S = \{a_3a_0, a_2a_0, a_1a_0, a_0\} + \{b_3b_0, b_2b_0, b_1b_0, b_0\}$$

A half adder taking x and y as input gives: $\text{sum} = x \oplus y$, $\text{carry out} = xy$



Carry propagation



$$\text{and } S = \{c_4, s_3, s_2, s_1, s_0\}$$

Question 2:

A Ternary digit is the base-3 equivalent of a binary bit

A binary digit can take 2 values: 0 or 1

A ternary digit can take 3 values: 0, 1 or 2

Ternary value	Binary encoding (to b_i)
0	00
1	01
2	10
3	11

$\downarrow$ $x_0 = \{a_1, a_0\} = 01 \rightarrow \text{Ternary } 1$
 $\downarrow$ $x_1 = \{b_1, b_0\} = 10 \rightarrow \text{Ternary } 2$
 $\downarrow$ $x_2 = \{c_1, c_0\} = 00 \rightarrow \text{Ternary } 0$

and then the selector $x_3 = \{s_1, s_0\} = 01 \rightarrow \text{choose input 1 (Ternary)}$

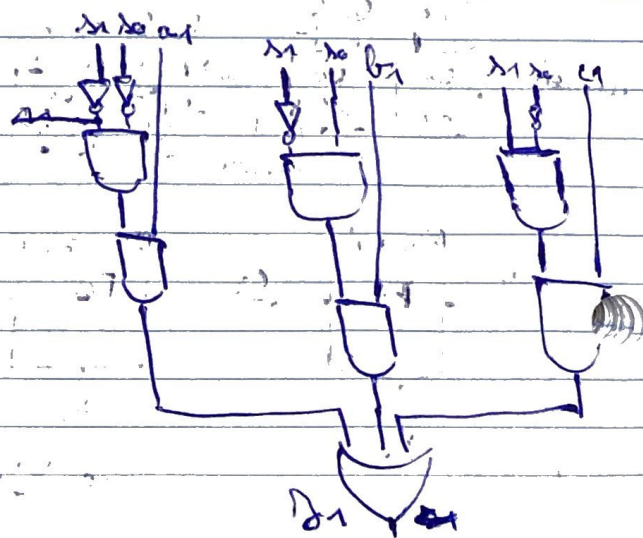
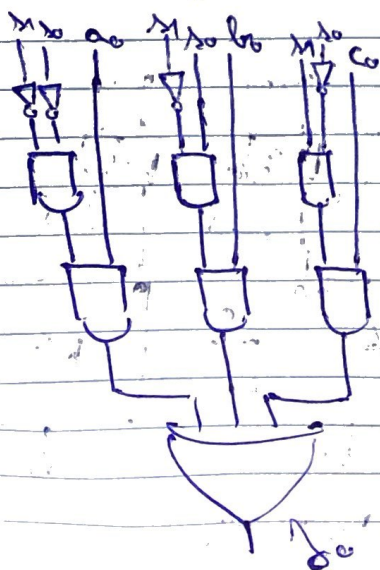
Then output $x_4 = \{y_1, y_0\} = x_1 = 10 \rightarrow \text{Ternary } 2$

~~Truth Table~~ ~~from E_1 to b_1, b_0 and a_1, a_0~~

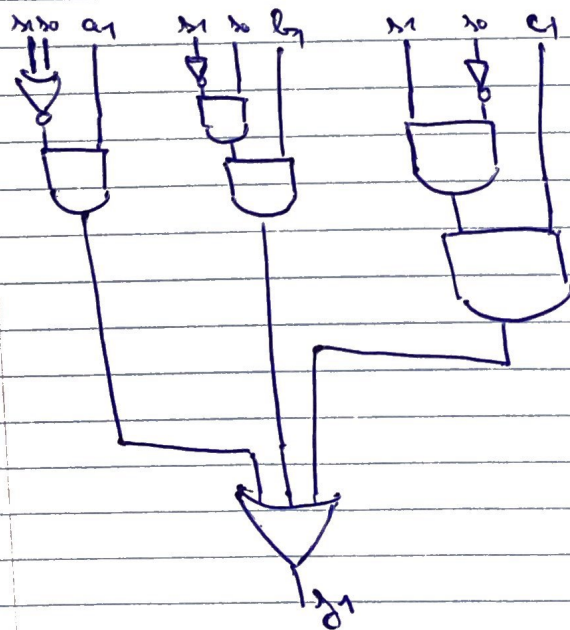
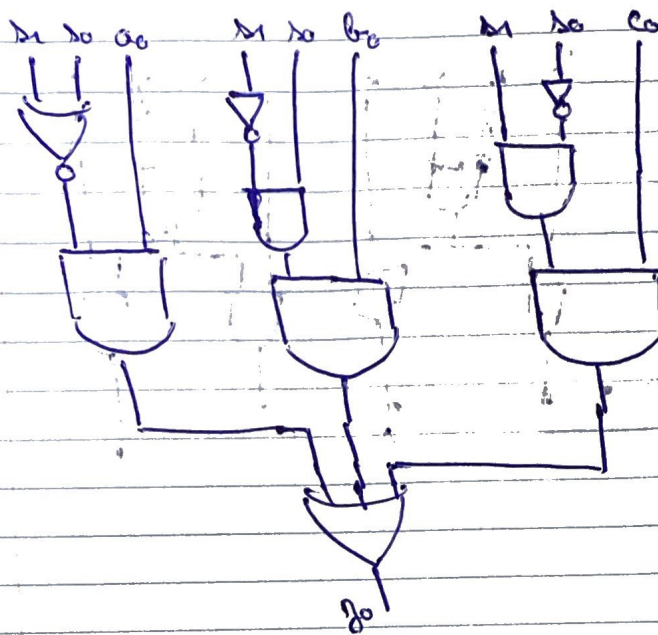
Hence let $E_0 = \overline{s_1} \overline{s_0}$ (select x_0)
 $E_1 = \overline{s_1} s_0$ (select x_1)
 $E_2 = s_1 \overline{s_0}$ (select x_2)

Then $y_1 = E_0 a_1 + E_1 b_1 + E_2 c_1$

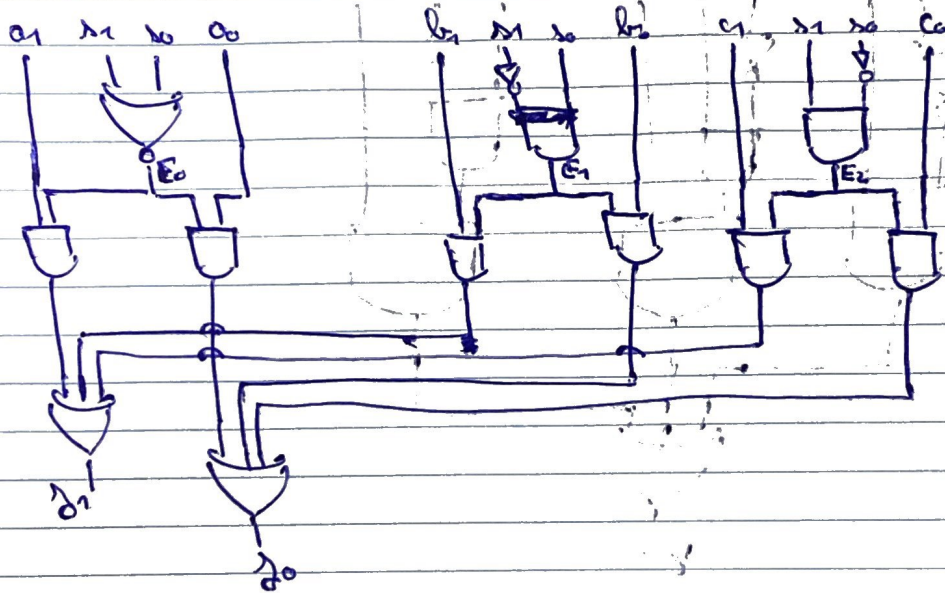
and $y_0 = E_0 a_0 + E_1 b_0 + E_2 c_0$



The circuit can be minimized by using Karnaugh maps.



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Question 3 (20%)

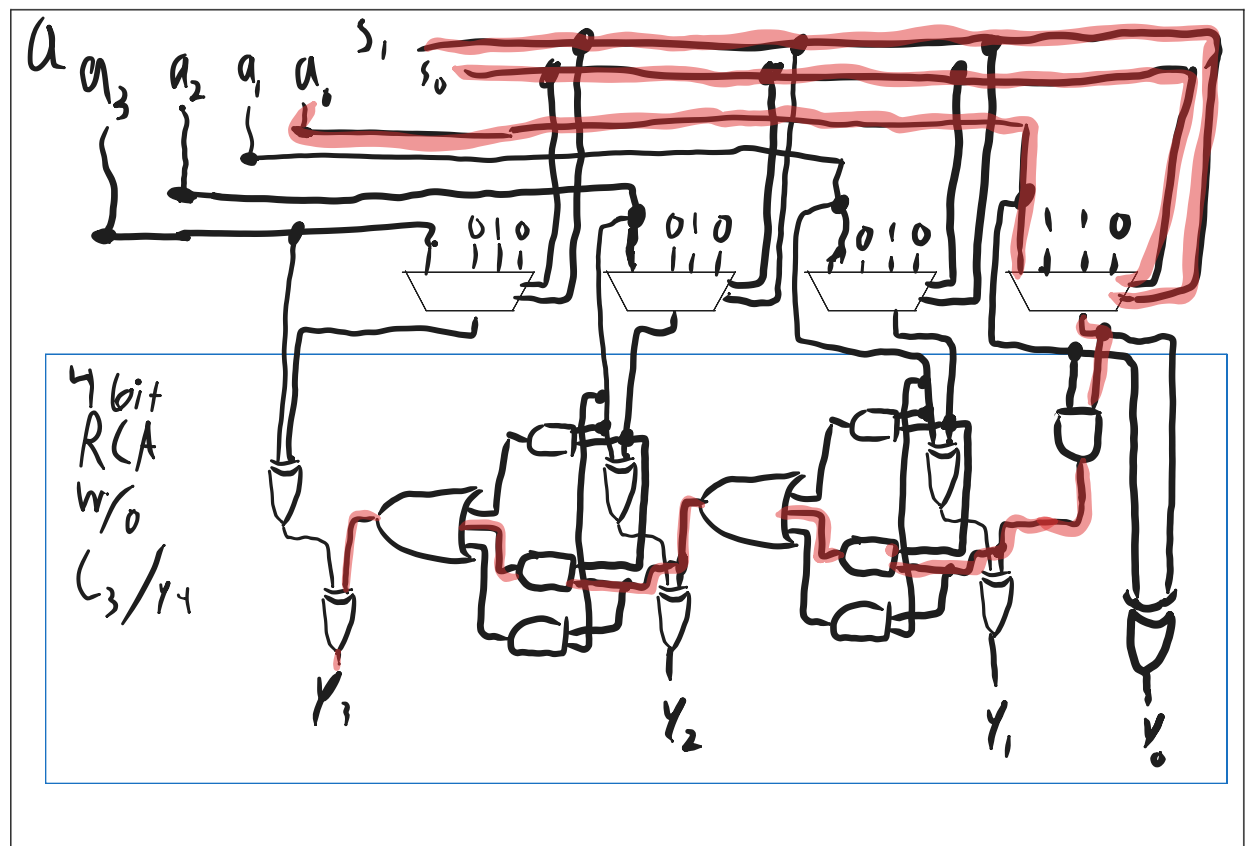
(a) Design and implement the following system:

Inputs: $X = x_3x_2x_1x_0$ and $S = s_1s_0$ Output: $Y = y_3y_2y_1y_0$

The functionality of the system is described in the table below. You can ignore any overflow that may occur. You can only use primitive gates, full-adders, half-adders, and 2-1 and 4-1 multiplexers. Any multiplexer can be considered a single gate, no need to implement. Alternatively, provide the internal circuitry of the half- and full-adders, these cannot be considered as a single gate. Hint: To implement a multiplication by two, consider shifting the output bits.

s_1s_0	Action
00	$Y = X$
01	$Y = X - 1$
10	$Y = X + 1$
11	$Y = 2 \cdot X$ (multiply by two)

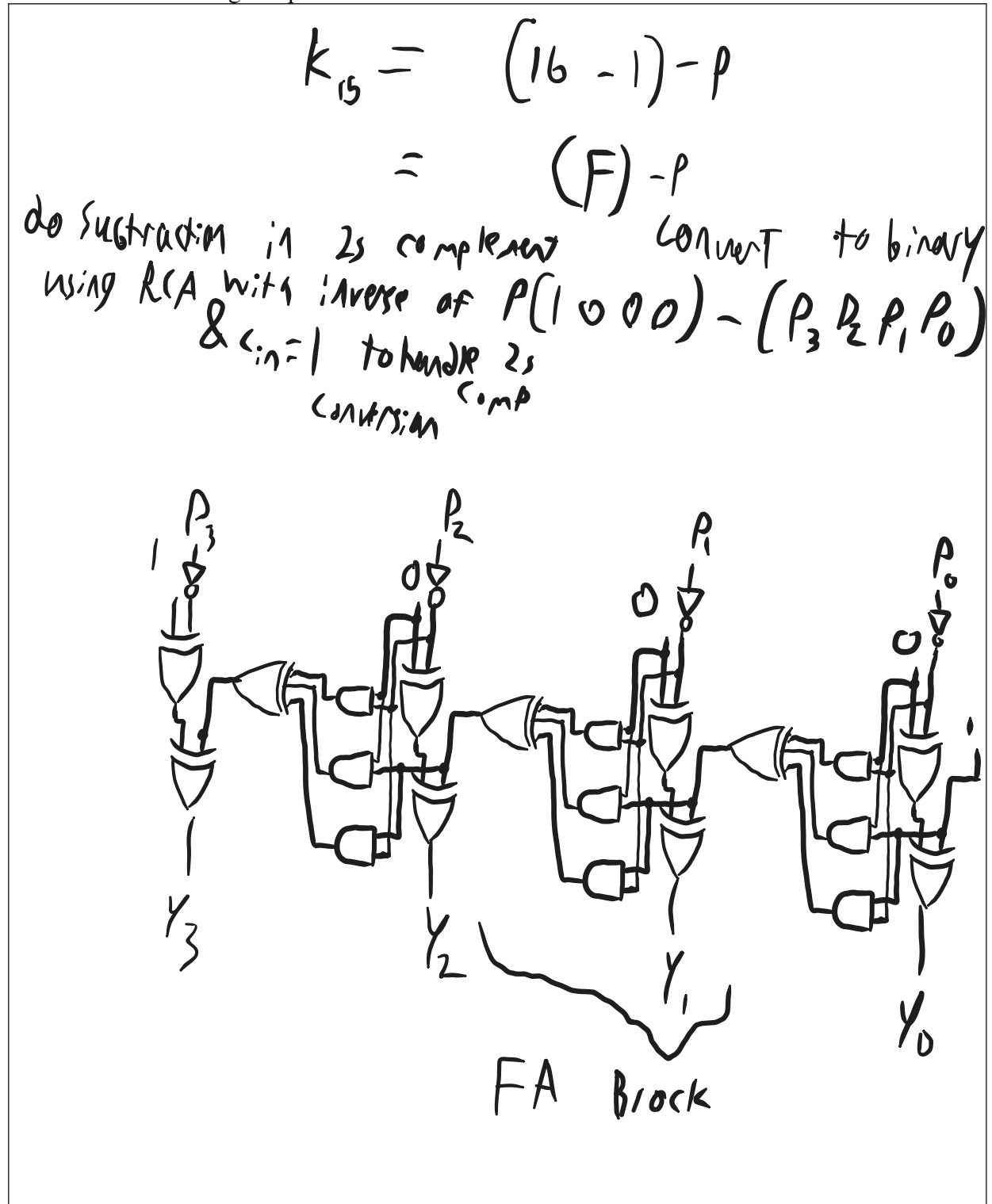
(b) Mark the critical path on the circuit you designed. What is the delay of the critical path? Assume that delay of a path is defined as the number of gates on the path. Also, assume that a multiplexer has the same delay as other primitive gates (this assumption is not true in most realistic cases and is only provided here to simplify the analysis).



b Critical path marked in red
passes through 7 gates, 1 4-1 mux,
3 and
2 or
1 XOR

Question 4 (15%)

Design and implement a circuit that generates the (15)'s complement of a HEX digit $p(p_3p_2p_1p_0)$. Provide all of the design steps.



Question 5 (15%)

Suppose that we want to determine how many of the bits in an eight-bit unsigned number are equal to 1. Design the minimum-cost circuit that can accomplish this task using only half- and full-adders.

